Java Full Stack Development — Training Material

> **One material policy for the Java program.** Java / Spring Boot notes live in this folder.
> **Shared topics** (same files as Python FS): `../FRONTEND/` (HTML/CSS/JS + `REACT/`), `../MYSQL/README.md`. Pointers only: `08_Frontend/`, `11_React/`. JPA/REST home is `09_Spring_Boot/` — `10_JPA_and_REST/` is a pointer, not a second copy.

Source syllabus: KSR Datavizon *Java Full Stack Development*
Audience: Learners preparing for entry-level Java / Spring Boot / full-stack roles
Default schedule: 20 weeks × 1 hour 15 minutes/day (Mon–Fri live) — see Session Calendar
Live contact = 125 hours (100 sessions) + daily homework
Stack: JDK 17 · Maven 3.9+ · MySQL 8+ · Spring Boot 3.x (Jakarta) · HTML/CSS/JS · React · Git/GitHub · VS Code or IntelliJ Community · Postman · JUnit 5

Start here

1. **Session Calendar** — 100 sessions, one focus each
2. **Tested Toolchain** · **Project deliverables** · **Capstone one-pager**
3. **Lab Assets** · **Biweekly Assessments**
4. Course Contents (syllabus map)
5. Open the **same** module files in Session Calendar order
6. Optional: AI Integration Lite — stretch after a deployed MVP only

Typical class shape (75 min)

Minutes	Block
5	Open lab kit
25	Concept demo / notes
35	Lab
10	Exit + homework

Teaching order

Schedule authority: always `Session_Calendar_100.md` (order below matches it).

Order	Path	Topic
1	01_Foundations/01_Computer_and_Programming_Foundations.md	Computer & programming foundations
2	02_Core_Java/01_Java_Setup_and_First_Programs.md	JDK setup & first programs

Order	Path	Topic
3	02_Core_Java/02_Variables_Datatypes_Operators_IO.md	Variables, types, operators, I/O
4	02_Core_Java/03_Flow_Control.md	Flow control
5	02_Core_Java/04_Strings.md	Strings
6	02_Core_Java/05_Arrays.md	Arrays
7	02_Core_Java/06_Methods.md	Static methods & overloading
8	03_OOP_Java/03_OOP.md	OOP (index) → Part 1 · Part 2 · Part 3
9	04_Collections_Exceptions_IO/01_Collections.md	Collections (`List` / `Set` / `Map`)
10	04_Collections_Exceptions_IO/02_Exception_Handling.md	Exception handling
11	04_Collections_Exceptions_IO/03_File_Handling.md	Files & I/O
12	05_Git_GitHub/01_Git_and_GitHub.md	Git & GitHub (Java `.gitignore`, Maven)
13	06_Database_MySQL/01_MySQL_for_Full_Stack_Developers.md	MySQL bridge + required labs — map: `MYSQL/README.md`
14	07_JDBC/01_JDBC_CRUD.md	JDBC CRUD (`PreparedStatement`)
15	../FRONTEND/01_HTML.md	HTML *(shared)*
16	../FRONTEND/02_CSS_and_Responsive.md	CSS & responsive *(shared)*
17	../FRONTEND/03_JavaScript_DOM_Forms.md	JavaScript, DOM, forms *(shared)*
18	09_Spring_Boot/01_Maven_and_Spring_Boot_Setup.md	Maven + Spring Boot setup
19	09_Spring_Boot/02_Spring_Web_REST.md	Spring Web REST (Thymeleaf overview only)
20	09_Spring_Boot/03_Spring_Data_JPA_Validation.md	Spring Data JPA + validation
21	09_Spring_Boot/04_REST_DTOs_Postman.md	REST, DTOs, Postman
22	09_Spring_Boot/05_Spring_Security_Lite.md	Spring Security lite (JWT)
23	09_Spring_Boot/06_Lab_CRUD_Auth_Consolidation.md	**Protected lab:** CRUD + auth
24	09_Spring_Boot/07_Lab_Deploy_Capstone_Debug.md	**Protected lab:** deploy + debug

Order	Path	Topic
25	09_Spring_Boot/08_Servlet_vs_Spring_Interview_Map.md	Servlet vs Spring (interview map only)
26	../FRONTEND/REACT/01_React_Fundamentals.md	React fundamentals *(shared)*
27	../FRONTEND/REACT/02_React_Hooks_Routing_API_Deploy.md	Hooks, routing, API, deploy *(shared)*
28	12_Testing_Clean_Code/01_Testing_Debugging_Clean_Code.md	Testing, debugging, clean code
29	13_Placement/01_Career_Resume_Interview.md	Resume, interview, placement
30	12_AI_Integration/01_AI_Integration_Lite.md	AI lite — **optional stretch only**

Pointer (not a second syllabus): 10_JPA_and_REST/ → teach JPA/REST from `09_Spring_Boot/` `03`–`04`.

Authored so far: Foundations + Core Java (`01`–`06`) + OOP Parts 1–3 + Collections + Exception Handling + File Handling + Git and GitHub + MySQL for Full Stack Developers (bridge; SQL depth in `MYSQL/`) + JDBC CRUD (`PreparedStatement`) + Maven and Spring Boot Setup + Spring Web REST + Spring Data JPA and Validation + REST, DTOs, and Postman + Spring Security Lite with lab kits 01–19 (kit 19: JWT register/login + protect POST; Lab 06 freeze 400/404 + demo bundle; Lab 07 Actuator health + env / PaaS / M0) and BW01 / BW02 / BW03 / BW04 / BW05 / BW06. Shared FRONTEND HTML/CSS/JS is already used. Protected labs `06_Lab_CRUD_Auth_Consolidation.md` and `07_Lab_Deploy_Capstone_Debug.md` (kit 19) and `08_Servlet_vs_Spring_Interview_Map.md` (kit 19 point-only; five comparison bullets) are authored. Later rows (CORS recap via `04`, React onward) are still planned file names until those phases are written.

What each module contains

Every teaching module follows the same shape:

- **Module Outcome** (3–5 can-do bullets) right after the header
- **Definition** / **Purpose** before syntax
- Important concepts with **Try these (2 programs)** + runnable Java (JDK 17); MySQL bridge Try these are SQL (depth in `MYSQL/`)
- Step-by-step annotated code
- Practice exercises
- **Mini-Project (Assignment)** — take-home homework with lab kit, Stretch, and **Done when** (when present)
- Interview Q&A
- Graded checks: `16_Assessments/` (biweekly) — not an Assessment block in every module

SQL depth lives in repo `MYSQL/` (required labs listed in the MySQL bridge module) — not duplicated here.

> Mermaid diagrams render on GitHub, many Markdown previewers, and Cursor. PDF builds pre-render them to SVG (`./scripts/build.sh java` needs Node/`npx` + Chrome). In plain text viewers you still see the diagram source as a readable sketch.

Sequence improvements vs brochure

Brochure order	Improved order (this material)	Why
Spring views before frontend	HTML/CSS/JS **before** Spring	Markup/JS before JSON or Thymeleaf overview
Git not explicit	Git early (Week 9, then continuous)	Portfolio habit for hiring
JPA without JDBC	**JDBC first**, then JPA	Learners see SQL before the ORM
Servlets/JSP as backbone	One **interview map** session only	Job market is Spring Boot + REST
React after “some backend”	After Spring REST + labs	SPA needs a real Java API
AI Integration last	**Stretch after** deployed MVP	Never delays core competence / placement
Thin Spring calendar	5 teach modules + **2 lab days** (`06`–`07`)	Career-switcher consolidation time

Brochure modules **preserved:** Core Java, OOP, Collections, Exceptions, JDBC, JPA, Spring Boot, REST, Security overview, Frontend, React, Git, MySQL, testing, Agile/resume/live project.

Maintainers

PDFs / optional share mirror (from **repo root**):

```
./scripts/build.sh java                # → PDF/JAVA/  (same notes as class)
./scripts/pack_student_share.sh java   # optional zip mirror; excludes private folders only
```

Python FS is unchanged: `./scripts/build.sh python` · `./scripts/pack_student_share.sh`.

Share habit: send each module PDF after that module is taught. Rebuilt PDFs under `PDF/` are committed and pushed to origin with the notes.
Private ops (keys, rubrics): `14_Trainer_Only/`.